

From Exact Numbers to Acquired Evidence

WikiGraphRAG v16: reader preparation and feasibility checkpoint

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Working addendum | preparation and authored controls | no natural QA evaluation

Abstract

Exact numeric decoding does not establish that a system selected the correct report, period, entity or population. This development checkpoint prepares the evidence and interface needed to test that distinction. A lossless storage amendment completes table extraction after a retained output-cap failure. The original fixed selection fills 13 of 24 planned table slots. A post-inspection visibility amendment fills 21 slots and covers all twelve filings, while retaining three unfilled slots and the original outcome. An acquired-only binding interface passes 71 authored checks after documented fixes. The native reader gate stops before model launch. These are preparation outcomes, not natural question-answering gains, retrieval improvements or a new algorithm. Amended taxonomy and source-render outcomes remain draft fields until their records are terminal.

1. What this checkpoint establishes

The research objective remains an automatic reader that can interpret a separately written request and select every required source binding. The previous typed numeric work supplied exact values at explicitly provided aspects; it did not solve that mapping problem. This checkpoint therefore builds source views, a bounded candidate pool and a common acquired-evidence interface before attempting a new retrieval policy.

All physical sources are the same known development filings. Table and question selection can be informed by source review, but that makes the eventual experiment an oracle-evidence competence diagnostic. It must not be described as retrieval success or held-out generalization. The draft protocol separates deterministic lexical filtering (B0), a model proposer with deterministic typed rendering (B1), and a direct model reader (B2).

Artifact	Observed outcome	Meaning
Table extraction	Complete after storage amendment	All fixed sources retained; DOM structure only.
Fixed source pool	21 / 24 table slots	Original outcome retained; source coverage restored, slots still unfilled.
Binding interface	71 authored checks pass	Tested access and schema behavior; not question interpretation.
Natural reader study	Not executed	No reader or retrieval performance conclusion.

Table 1. Preparation outcomes are distinct from the later competence experiment.

Preparation improves source coverage but still does not supply admitted questions or a competent reader. The visibility-only selection amendment is explicitly post-inspection development; it does not retrospectively make the original rule successful. The native candidate has not produced an evaluated answer.

2. Complete structural views, with the failed attempt retained

The first extraction hits its fixed storage cap and completes nine of twelve sources; three source failures, partial outputs and hashes remain recorded. A separate amendment changes storage to gzip without changing extraction logic. It completes the cohort and preserves the earlier output through a lossless compressed archive. Neither compression nor successful serialization repairs the original failed execution.

The views retain complete table rows and cells, captions, neighboring paragraph references, declared spans, source anchors, numeric occurrence links and recognized hiding cues. The extraction records DOM ownership rather than a browser-computed cell grid. It does not establish visual alignment, CSS visibility, semantic header attachment or complete supporting footnotes.

Quantity	Original attempt	Lossless amendment
Physical-source denominator	12	12
Numeric occurrences in completed sources	17278	23651
Tables in completed sources	1340	1765
Rows in completed sources	12619	16861
External artifact bytes	99934266	10768154

Table 2. Original totals describe completed sources only; the fixed source and occurrence denominators are retained in both receipts. The amended payload is compressed, so byte counts are not parsing-speed measurements.

A separate streaming audit checks the compressed containers, source/fact coverage and aggregate identities. Its own first attempt incorrectly looked for an outside-table record instead of the source-level ordinal field and flagged all sources. The corrected audit preserves that failed audit and passes its checks without changing the data. These are provenance and accounting checks, not independent semantic annotations.

An authored rendering check exposed a separate problem: Story passed its code tests but placed merged year headings over the wrong numeric columns. A LibreOffice conversion of the same HTML retained the tested year/value geometry, but lost declared borders and width and used office-specific layout. It is therefore a qualified inspection aid, not evidence of browser fidelity. Per-source visual and CSS review is still required before admitting a question or reference.

3. A retained coverage gap and a visibility-only amendment

The original selection is deliberately narrow: leaf tables, complete row/cell ownership, no recognized hiding cue, a bounded numeric occurrence count, multiple resolved duration periods, a revenue or net-sales phrase within one row, and a bounded complete DOM text length. Nil, unsupported and invalid occurrences remain in the occurrence count. Binding-resolved period presence is separate from numeric usability.

The original rule selects tables from only part of the source cohort. A descriptive replay identifies candidates rejected solely because recognized hiding cues occur on empty layout subtrees. A separately frozen amendment permits such cues only on strict descendants meeting conservative emptiness checks; cues on the table or an ancestor, content-bearing subtrees and id/class hooks remain blocked. No row, value or source is removed or rewritten, and every nonvisibility predicate is replayed against the parent outcome.

Physical filing	Original	Amended	Unfilled
PFE_2023.html	2	2	0
PFE_2024.html	2	2	0
NFLX_2022.html	0	2	0
NFLX_2023.html	0	2	0
NVDA_2022.html	2	2	0
NVDA_2023.html	2	2	0
LLY_2023.html	1	2	0
LLY_2024.html	2	2	0
BRK-B_2022.html	1	1	1
BRK-B_2023.html	0	2	0
SLB_2022.html	1	1	1
SLB_2023.html	0	1	1

Table 3. Selected table counts under the original and amended rules; unfilled refers to the amended rule. Both use the first two eligible tables in XML order. Filenames identify physical objects, not verified fiscal periods; the Berkshire filename/period mismatch remains recorded.

The amended rule makes 29 more tables eligible and covers every source, but still leaves three planned table slots empty. Its unchanged first-two rule retains seven earlier selections and displaces six with earlier eligible tables; every original packet is preserved. This is a post-inspection engineering repair, not a pristine confirmatory result or proof of rendered visibility.

A separate pre-freeze code review also corrected a regex that could span two rows. Both corrections have explicit timing and provenance. Table slots are not question counts: neither selection produces a question, gold binding, complete evidence pack or reader prediction.

4. Acquired-only binding and a common evidence budget

The adapter stores only an immutable acquired evidence pack. Each fact has an opaque occurrence handle, its exact value or null, visible labels, and six explicit bindings: concept, entity, period, unit, population and physical source. Each available binding cites observed witnesses; an unavailable binding remains null. The adapter has no full-document callback or hidden-source lookup. Pack construction must still remove every duplicate disclosure of withheld evidence from visible text and labels.

B1 proposes a bounded beam of joint claim assignments and declares unresolved interpretation explicitly. The finalizer scans all exposed compatible occurrences, coalesces equal scoped quantities while retaining every occurrence and witness, and blocks unsupported or conflicting possible matches. Equal values in different years, units, entities or source versions remain distinct scoped claims. A compact shared-registry projection is lossless and identical for B1 and B2; unused registry content is rejected rather than silently discarded.

B0 uses fixed token overlap and witnessed constraints, with no question-specific mappings. B2 uses the same evidence and supplies quantities directly. Wrong canonical quantities remain in its output and are graded; a refusal prefix does not exempt a material claim. Syntax and exposed support do not certify that an answer satisfies the question.

Authored audit outcome	Initial review	Final combined run
Tests run	28	71
Failures	6	0
Errors	1	0
Skipped	0	0

Table 4. The initial run is the separate reviewer suite; the final run includes both the owner and expanded reviewer suites. These counts are engineering tests, not examples of natural QA.

The independent review found duplicate-joint assignment/provenance errors, ignored compact registry content and an uncontrolled Unicode error. Those failures were saved before correction. The final code passes the expanded controls and preserves source/citation identity. No automatic question-to-fact performance claim follows from these tests. Implementation and review used separate model-agent roles; these controls are not human annotations or independent training populations.

5. Execution gates and scope of evidence

Dependency capture records 117 resources and one transport timeout, leaving eleven complete source closures. Original validation launches no worker. A separate small-process amendment replaces the total-current soft cap with a conservative pressure proxy and reserve under the same kernel hard limit; it does not satisfy the old cap. That attempt fails on all sources: mixed physical/logical DOM traversal prevents completed inventories, and namespace PID lookup makes its RSS reports unreliable. No RSS-enforcement claim is made for that attempt. A separately reviewed repair preserves source bytes, comparison logic and processor options.

The pinned Qwen3.5-9B UD-IQ3_XXS candidate was acquired with its exact file hash, but its authored native gate stopped before launch because observed total memory exceeded the unchanged guard. The receipt labels this tokenization_failed; the follow-up review establishes that no model process, prompt tokenization or completion occurred. This is a feasibility failure, not a reading error, and it supplies no stronger-reader evidence. No retry or guard relaxation was made within this attempt.

Processor completion, numeric agreement and source identity are separate claims. Candidate taxonomy metadata does not establish original SEC byte identity; pinned options exclude SEC filing-rule, calculation and unit-registry checks. Rendering likewise requires actual visual/CSS review before source admission.

Authored native gate	Observed
Planned B1/B2 completions	8
Native model processes started	0
Prompts tokenized	0
Completion calls started	0
Observed total memory (bytes)	8471801856
Frozen total-memory guard (bytes)	7784628224

Table 5. The resource guard stopped the attempt before model launch. All planned request states remain not attempted; no model performance measurement was obtained.

Gate	State	Claim allowed
Original taxonomy run	12 prelaunch refusals	No completed inventory or comparison.
Resource amendment	12 processor failures	RSS enforcement not established.
DOM/RSS repair	Draft: terminal evidence required	No amended result asserted.
Natural source rendering	Draft: terminal evidence required	No browser-fidelity or admission claim.

Table 6. Every attempt is retained. Resource-policy changes apply only to the small processor and office-conversion jobs; the native reader is not retried.

Draft checkpoint: terminal amended taxonomy and natural source-render receipts remain required before final publication. The original native resource failure is final for this attempt; no native model retry is part of the small-process resource amendments.

6. Research decision and reproducibility

This checkpoint does not yet justify spending the next experiment on a novel retrieval policy. The necessary intermediate result is evidence that an automatic reader can correctly bind separately authored requests when complete evidence is already supplied. That requires reviewed visible source support, frozen question/reference sets, valid missing-evidence controls, and complete accounting of planned cells.

Both pool outcomes remain part of the record: the original source-coverage gap and the amended cohort coverage with residual empty table slots. The amendment does not establish that a question can be answered from each selected block. Later question/evidence admission must inspect actual visible dependencies. Reader tuning must likewise preserve the first result and record a revised model, prompt or interface as a new attempt.

The common interface provides a concrete engineering foundation for testing retrieval later. It distinguishes missing binding evidence from question ambiguity and preserves exact scoped alternatives. Those distinctions are conventional prerequisites. They become a research contribution only if an independently designed method improves a meaningful task under matched source access, reader strength and full cost accounting.

Public artifact	Purpose
table_view_inventory_v16 / v16_1	Retained failure and complete compressed amendment.
table_storage_independent_audit_v16	Container, source-coverage and aggregate audit.
reader_pool_v16 / v16_1 + protocols	Original and amended selection, empty slots and exact rules.
reader_binding_code_review_v16	Initial authored failures and corrected interface controls.
reader_preparation_study_v16 + render manifest	Canonical text, evidence hashes and checked numeric table cells.

Table 7. Filenames are rooted in results/, data/reader_binding_v16/ or paper/ as appropriate. Full paths and SHA-256 identities are provided in the canonical manuscript and render manifest.

The published package includes source and tests, fixed rules and protocols, aggregate receipts, the claim ledger and editable generic TeX. Source-bearing views, detailed typed records, renderer outputs and model prompts/outputs remain external where redistribution has not been established. The PDF is produced with ReportLab and visually checked; the generic TeX has not been compiled or verified against the ACL template.